Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous Institute Affiliated to University of Mumbai)

ESE Examination
December 2025

Max. Marks: 100
Class: T.E. Computer Engineering
Course Code: CE302/CSE302
Name of the Course: Software Engineering

Duration: 3Hrs.
Semester-V
Program: B.Tech.(CE)

Instructions:
(1) Write questions in sequence and start new question on new page.
(2) Assume suitable data if necessary.

Course Outcomes: At the end of the course student will be able to:
(1) **Illustrates** software development lifecycle models.
(2) **Analyze** user requirements and specifications to identify functional and nonfunctional requirements of a software system.
(3) **Design** structured and object-oriented system using appropriate design patterns.
(4) **Apply** project management principles to schedule tasks, estimate development costs, and manage risks in software system.
(5) **Apply** software testing methods to create test cases that validate system functionality.

<table>
<tr><th>Q. No.</th><th></th><th>Max marks</th><th>CO-BL</th></tr>
<tr><td>Q.1 A)</td><td>Distinguish between functional and non-functional requirement. For a ride sharing application like Uber/Ola, write at least 8 user stories for the app. Apply Agile framework to develop and deliver the application.
OR
What is the importance of process model in development of software system? Identify 4 challenges faced during requirement gathering and techniques to resolve the challenges. Write functional and non-functional requirements for an "Automatic Question Paper Generator" application.</td><td>10</td><td>CO1,2-3</td></tr>
<tr><td>Q.1 B)</td><td>Use PERT analysis method to draw the network diagram, find the critical path and estimated time of completion for the project using data in the table below.
<table>
<tr><th>Activities</th><th>Immediate Predecessor</th><th>Optimistic Time</th><th>Most likely Time</th><th>Pessimistic Time</th></tr>
<tr><td>A</td><td>-</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>B</td><td>-</td><td>3</td><td>5</td><td>7</td></tr>
<tr><td>C</td><td>-</td><td>4</td><td>7</td><td>10</td></tr>
<tr><td>D</td><td>A</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>E</td><td>B</td><td>3</td><td>4</td><td>11</td></tr>
<tr><td>F</td><td>C</td><td>4</td><td>8</td><td>12</td></tr>
<tr><td>G</td><td>C</td><td>3</td><td>3</td><td>9</td></tr>
<tr><td>H</td><td>E,F</td><td>6</td><td>6</td><td>12</td></tr>
<tr><td>I</td><td>D</td><td>5</td><td>8</td><td>11</td></tr>
<tr><td>J</td><td>H,G</td><td>3</td><td>3</td><td>9</td></tr>
</table></td><td>10</td><td>CO4-3</td></tr>
</table>

Q.2 A)	Identify 5 risks for an ecommerce website development project and suggest initial mitigation strategies. Classify the following as Project Risk /Technical Risk/ Business Risk 1. Scope creep. 2. Change in government regulation. 3. Integration problems due to faulty design. 4. Hardware malfunction. 5. Team turnover.	10	CO4-3
Q.2 B)	Illustrate with suitable examples the different types of cohesion? OR Illustrate with suitable examples the different types of coupling?	10	CO3-2
Q.3 A)	What do you mean by balancing a DFD? Draw a labelled DFD up to level 2 for an entrance examination system. Also, write the data dictionary. In this system an applicant has to make the application by entering all the details and paying the fees for the application. The mode of payment is through credit/debit card or online. On the other hand, verification committee is responsible to verify each of the applications. The system has an enrolment office which takes care of generating and dispatching the admit cards. On receiving the admit card, an applicant is typically allowed to sit for the examination. The examiner checks the answer sheets and submits the marks to the system. Based on the marks given by the examiner, the enrolment office produces the grade card and dispatches those to the applicants.	10	CO2-3
Q.3 B)	A student wants to issue a book from library. At first the student requests the librarian for a book which he/she needs. The librarian searches for a book in the database. The book database system searches the book availability and checks the book availability and checks the details of the book. The database system returns the details of the book to the librarian. Then the librarian requests library card from the student. The student provides the library card, and the librarian updates the details into the database. After completion of entering the details in the database. A receipt is generated for the librarian. Finally, a book is issued to the students. Based on the details of the above scenario, Draw sequence and collaboration diagram.	10	CO3-3
Q.4 A)	A software development team is designing a real-time stock market monitoring system where multiple users should automatically receive updates whenever stock prices change. The system also needs to maintain a clear separation between the user interface, data, and business logic for	10	CO3-4

	ease of maintenance and scalability. 1. Identify which design patterns are most suitable to implement this system and justify your selection. 2. Compare how the MVC architectural pattern and the Observer design pattern can work together in this context to achieve loose coupling and dynamic data updates. 3. Illustrate with a brief diagram how both patterns can be integrated in the system design.		
Q.4 B)	Analyze how a software developer can modify the coupling between classes to improve maintainability. Illustrate your answer with an example demonstrating the change from high coupling to low coupling.	10	CO3-4
Q.5 A)	Justify path coverage provides stronger assurance than statement coverage in the given context.	10	CO5-5
Q. 5 B)	You are testing an online shopping application where the Payment module interacts with the Order Processing module and Inventory module. Design and perform Integration Testing for this scenario using any one of the following approaches: 1. Top-Down Integration 2. Bottom-Up Integration. 3. Sandwich (Hybrid) Integration Explain the steps involved, test data selection, and expected outcomes.	10	CO5-3

****